

1. The result displayed from `console.log(name)` is?



```
let name = 'Dedi'
name += 'Kusnandar'
name = 'Ciro Alves'
console.log(name)
```

- a. Dedi
- b. Kusnandar
- c. Ciro Alves
- d. Dedi Kusnandar Ciro Alves

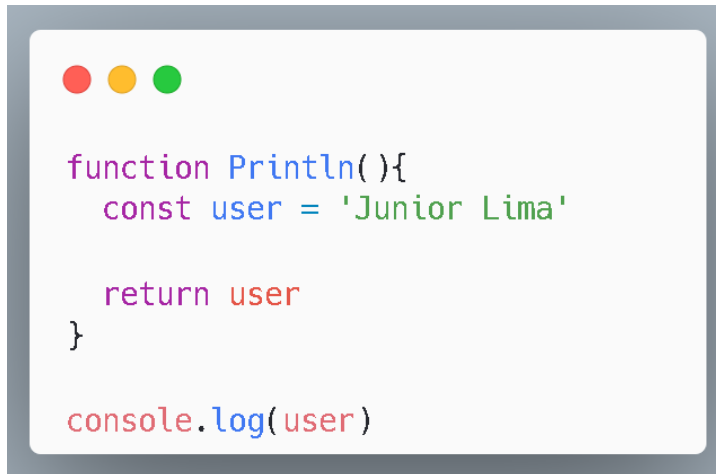
2. LIFO (Last In First Out) is a concept used in one type of data structure. The data structure that uses LIFO is:

- a. Stack
- b. Queue
- c. Hash Map/Hash Table
- d. LinkedList

3. Which of the following is a characteristic of a queue data structure?

- a. Last-In, First-Out (LIFO)
- b. First-In, First-Out (FIFO)
- c. Random Access
- d. Sorted Order

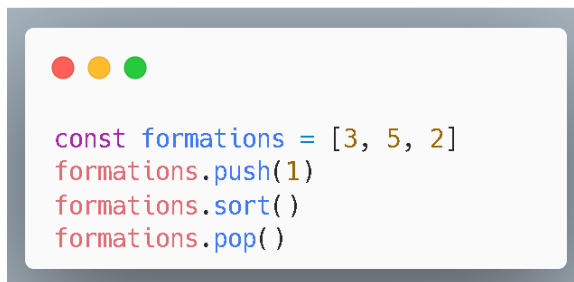
4. What will be displayed by console.log if the code below is executed?



```
function Println(){  
    const user = 'Junior Lima'  
  
    return user  
}  
  
console.log(user)
```

- a. Undefined
- b. Null
- c. Junior Lima
- d. `ReferenceError` user is not defined

5. What the final value of variable formations?



```
const formations = [3, 5, 2]  
formations.push(1)  
formations.sort()  
formations.pop()
```

- a. [1, 2, 3]
- b. [2, 3, 5]
- c. [5, 2, 1]
- d. [1, 2, 5]

6. What is current value of variable result?



```
let result = 'Hello'

if('0' !== 0) result += ', World!'
```

- a. 'Hello'
- b. ', World!'
- c. 'Hello, World!'
- d. TypeError: Assignment to constant variable

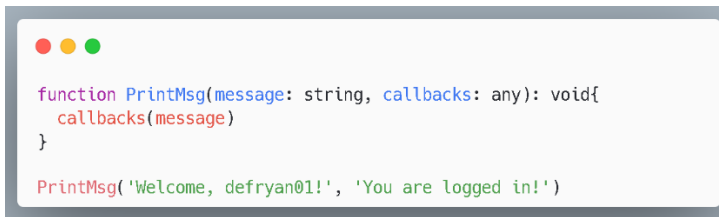
7. The following are correct statements regarding Array in JavaScript, except?

- a. Has `.length`` method
- b. It can accommodate more than one value
- c. It can hold all data types and values in javascript, including themselves
- d. It can hold all data types and values in javascript except for object and function

8. The difference between `while` and `do while` in javascript is:

- a. In a while loop, the loop will only run if the given condition is false, whereas in a do while loop, it will continue to run continuously even if the given condition is true
- b. In a while loop, the condition is checked after each iteration before executing the statements inside the while loop, whereas in a do while loop, the condition is checked first before each iteration
- c. A while loop will only run if the given condition is true, whereas a do while loop will continue to run continuously even if the given condition is false
- d. In a while loop, the condition is checked before each iteration before executing the statements inside the while loop, whereas in a do while loop, the condition is checked after the iteration has run

9. What will happen if the following program is executed?



```
function PrintMsg(message: string, callbacks: any): void{
  callbacks(message)
}

PrintMsg('Welcome, defryan01!', 'You are logged in!')
```

- a. `Welcome, defryan01`
- b. `You are logged in!`
- c. `TypeError: callback is not a function`
- d. `ReferenceError: callback is not defined`

10. The result of the code bellow when executed is:

```
const name: string = 'Indivara'
const age: number = 17

if(!name || !umur) console.log('Data not valid')
else if(name && umur > 17) console.log('Verification success!')
else if(name && umur < 17) console.log('Verification failed!')
else console.log('Verification on process!')
```

- a. Data not valid
- b. Verification success!
- c. Verification failed!
- d. Verification on process!

11. What result would be displayed if we execute the command `console.log(nmbr)` from the variable data in the last line, and what is its data type, `typeof nmbr` ?

```
let nmbr: number | string = '12z'
nmbr = Number(nmbr)
nmbr = String(nmbr)
```

- a. NaN and String
- b. '12' and String
- c. NaN and Number
- d. 12 and Number

12. What the result of console bellow?



```
const number = 'WD123'  
console.log(parseInt(number))
```

- a. NaN
- b. 123
- c. '123'
- d. NaNNaN123

13. What the result of console bellow?



```
const number = '1W2D3'  
console.log(parseInt(number))
```

- a. 1
- b. 1NaN2NaN3
- c. '1NaN2NaN3'
- d. '1'

14. The difference between `forEach()` dan `map()` is:

- a. `forEach()` can be used to loop through objects just like a for-loop
- b. `map()` is used if you only need a loop without creating a new array
- c. A `forEach()` loop can be stopped with `break`
- d. `forEach()` does not produce a return or a new array, whereas `map()` produces a new array

15. What is the output of the program below:



```
console.log(String(12) == '12')
```

- a. true
- b. false
- c. `SyntaxError`
- d. `ReferenceError`

16. Consider the following code, Based on the code, which time complexity best describes the code?

```
function checkDuplicate(arr: any[]): void {  
  for (let i = 0; i < arr.length; i++) {  
    for (let j = i + 1; j < arr.length; j++) {  
      if (arr[i] == arr[j]) return true;  
    }  
  }  
  return false;  
}  
  
console.log(checkDuplicate([1, 2, 3, 1]));
```

- a. $O(n)$
- b. $O(1)$
- c. $O(n^2)$
- d. $O(\log n)$

17. Which of the following is NOT a primitive data type in JavaScript?

- a. { fullname: "Immanuel" }
- b. True
- c. "Hello, World!"
- d. 125

18. Which of the following is NOT a valid way to declare a variable in JavaScript/Typescript?

- a. `int point = 100;`
- b. `const text: string = "404"`
- c. `let isMarried = true;`
- d. `var title: string = 101;`

19. What does the var keyword do in JavaScript?

- a. Declare function scoped variable
- b. Declare variable with constant value
- c. Declare a block-scoped variable
- d. Declare global-scoped variable

20. How to access nested property object safely?

- a. Using the ! operator
- b. Using the ?. operator
- c. Using the ?? operator
- d. Using the == operator